

Solution

Brac University
CSE427: Machine Learning
Spring 2026
Quiz 02
Set B

Duration: 25 minutes

Name:	ID:	Section: 05
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Question 1:

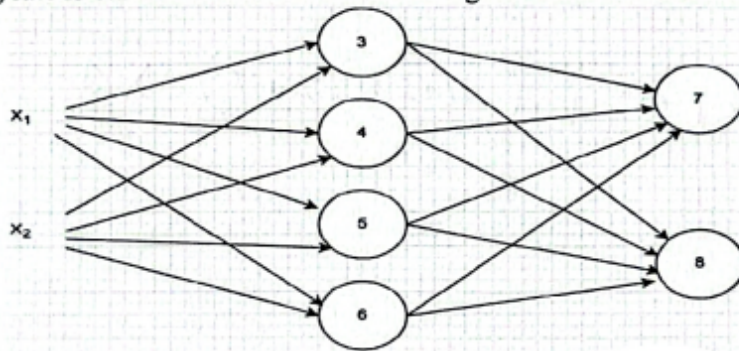
In the following neural network find the updated weights of W_{23} and W_{37} . Initially the weights of all the links are 1.5 and the biases of node 3, 4, 5 and 6 are -1 each. Assume the biases of node 7 and 8 are -3.0 and -2.5 respectively. Assume that the input is $X_1 = 1.5$ and $X_2 = 1.25$ and the target output is $Y_7 = 1$ and $Y_8 = 0$. The learning rate is 0.1 and all the neurons are using the Relu activation function.

all weight = 1.5

$b_3 = b_4 = b_5 = -1$

$b_7 = -3$

$b_8 = -2.5$



$y_7 = 1$

$y_8 = 0$

We know,

$$W_{23} = W_{23} + \eta \delta_3 x_2$$

$$W_{37} = W_{37} + \eta \delta_7 Y_3$$

$$Y_3 = \text{ReLU}(W_{13} x_1 + W_{23} x_2 + b_3) = \text{ReLU}(1.5 \times 1.5 + 1.5 \times 1.25 - 3) = \text{ReLU}(3.125) = 3.125$$

$$Y_3 = Y_4 = Y_5 = Y_6 = 3.125 \quad [\text{as all weights are same}]$$

$$Y_7 = \text{ReLU}(W_{37} Y_3 + W_{47} Y_4 + b_7) = \text{ReLU}(1.5 \times 3.125 + 1.5 \times 3.125 - 3) = \text{ReLU}(15.75) = 15.75$$

$$Y_8 = \text{ReLU}(W_{38} Y_3 + W_{48} Y_4 + b_8) = \text{ReLU}(1.5 \times 3.125 + 1.5 \times 3.125 - 2.5) = \text{ReLU}(15.75) = 15.75$$

$$= \text{ReLU}(16.25) = 16.25$$

$$S_7 = \text{Err}_7 Y_7' = (O_{b7} - Y_7) Y_7' = (1 - 15.75) \times 1 = -14.75$$

$$S_8 = \text{Err}_8 Y_8' = (O_{b8} - Y_8) Y_8' = (0 - 16.25) \times 1 = -16.25$$

$$\begin{cases} \text{if } x > 0 & \text{ReLU}'(x) = 1 \\ x < 0 & \text{ReLU}'(x) = 0 \end{cases}$$

$$S_3 = (S_7 W_{37} + S_8 W_{38}) Y_3'$$

$$= ((-14.75) \times 1.5 + (-16.25) \times 1.5) \times 1$$

$$= -46.5$$

so,

$$W_{23} = W_{23} + \eta S_3 x_2$$

$$= 1.5 + 0.1 \times (-46.5) \times 3.25$$

$$= -4.3125$$

$$W_{37} = W_{37} + \eta S_7 Y_3$$

$$= 1.5 + 0.1 \times (-14.75) \times 3.125$$

$$= -3.109375$$